

Software Landscape

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ITAD Software Landscape

What commercial ITAD platforms reveal

Commercial ITAD platforms are useful as functional references because they show which workflows become important as an operation grows.

MAKOR ERP

MAKOR markets software specifically for electronics recycling and ITAD. Its advertised workflow areas include:

- CRM/leads and quotations
- contracts and service jobs
- pickups/logistics
- receiving
- serialisation and barcode tracking
- temporary storage/location tracking
- data security
- testing
- repair
- upgrading
- component harvesting
- teardown
- pallet/container tracking
- resale
- downstream disposition
- customer settlements
- certificates and reporting
- customer portal functions

For Dubbo E-Waste, MAKOR is more useful initially as a feature map than as something we necessarily need to buy.

RazorERP

RazorERP is another commercial platform oriented toward electronics recyclers and ITAD businesses. Particularly relevant capabilities include:

- inventory and asset tracking
- grading
- processing workflows
- fair-market-value/resale logic
- service contracts
- customer settlements
- marketplace integrations
- integrations with specialist sanitisation software

It is especially useful as a reference for the resale/value-recovery side of ITAD.

Phonecheck

Phonecheck is device-focused and demonstrates a useful record model:

- identify device
- capture serial/IMEI
- maintain chain of custody
- perform diagnostics
- perform/record sanitisation
- attach certification to the device record

That same record model can apply to laptops, desktops, drives and other data-bearing assets.

Open-source tools

Snipe-IT

Role: asset management.

Useful existing concepts:

- unique asset IDs
- serial numbers
- manufacturers/models
- asset status
- locations
- custom fields
- QR/barcode labels
- attachments
- REST API

Possible Dubbo E-Waste statuses:

Received
↓

Quarantine
↓
Data-bearing
↓
Wiping
↓
Tested
↓
Repair / Refurbish
↓
Reuse / Donate / Sell / Parts / Recycle
↓
Completed

Recommendation

Investigate Snipe-IT as either:

1. the initial asset database behind a custom front-end; or
2. a reference data model.

Avoid a deep fork unless necessary. API integration or a purpose-built database may remain cleaner.

GLPI

Role: IT asset management + service desk.

Relevant functions include:

- assets
- inventory
- tickets
- contracts
- software
- users
- locations
- lifecycle records
- APIs

GLPI may make sense if Dubbo E-Waste evolves into a combined:

ITAD + Refurbishment + Repair + IT Support

For a small pure e-waste operation it may be more complexity than needed.

OCS Inventory NG

Role: automated hardware/software discovery.

Potential workflow:

Scan asset QR
↓
Boot/agent identifies computer
↓

```
CPU / RAM / storage / NIC / OS / serial data collected
↓
Data automatically attached to asset
```

This can greatly reduce manual transcription during intake/testing.

smartmontools

Role: disk health and identity.

Useful information can include:

- manufacturer/model
- serial number
- capacity
- SMART status
- error history
- power-on hours
- temperature
- SSD/NVMe wear indicators

A custom interface could convert raw smartctl output into operator-friendly results.

Example:

```
Samsung 870 EVO 250 GB
Health: PASS
Power-on hours: 4,821
Errors: 0
Wear indicator: acceptable
Reuse route: eligible
```

The final reuse decision should be based on documented test policy rather than one SMART field alone.

Memtest86+

Role: RAM testing.

Could form part of a standard refurbishment test suite alongside:

- storage test
- battery health
- CPU/stability check
- display
- keyboard
- trackpad/mouse
- Wi-Fi/Ethernet
- USB
- audio

- webcam
- charging

Results should feed into the central device record.

FOG Project

Role: PXE deployment, imaging and batch processing.

Potential ITAD/refurbishment uses:

- PXE boot
- disk imaging
- operating-system deployment
- inventory collection
- scripted jobs
- disk tests
- memory tests
- wipe tasks
- multicast deployment

This becomes increasingly useful when processing batches of ex-business or ex-school computers.

nwipe and ShredOS

Role: data sanitisation.

nwipe is an open-source disk wiping tool. ShredOS provides a lightweight bootable environment built around nwipe.

Potential pilot use:

- wipe station
- multiple disk processing
- method selection
- verification
- log capture

Boundary

Dubbo E-Waste should build the evidence and workflow wrapper, not its own low-level erasure algorithm.

For higher-assurance enterprise/government ITAD, commercial products such as Blancco or other independently validated sanitisation tools may still be required depending on customer, contractual, regulatory and assurance requirements.

Functional comparison

Area	Commercial examples	Open-source starting point	Custom-build suitability
Receiving/intake	MAKOR, RazorERP	Snipe-IT / GLPI	High
Barcode/QR tracking	MAKOR, RazorERP	Snipe-IT	High
Chain of custody	ITAD platforms	Custom	Very high
Customer/job management	MAKOR, RazorERP	ERPNext / GLPI	High
Hardware discovery	ITAD diagnostics	OCS Inventory	High as integration
Storage diagnostics	Commercial suites	smartmontools	High as integration
Memory testing	Commercial suites	Memtest86+	High as integration
Imaging	ITAD tools	FOG	High as integration
Erasure	Blancco, BitRaser etc.	nwipe/Shred0S	Integrate, don't reinvent
Erasure certificates	Commercial tools	Logs + custom reporting	Very high
Repair workflow	MAKOR	GLPI/custom	High
Parts harvesting	MAKOR	Custom	Very high
Resale inventory	RazorERP, MAKOR	ERPNext/Snipe-IT	High
Marketplace integration	RazorERP	eBay APIs	High
Client settlements	RazorERP/MAKOR	ERPNext + custom	Medium
Weight/mass balance	Recycling ERPs	Custom	High
Downstream tracking	Recycling ERPs	Custom	Very high
Recycling certificates	Recycling ERPs	Custom PDF/reporting	Very high
Customer portal	MAKOR etc.	Custom	High

Bottom line

The specialist utilities are already available.

The gap worth building is a lightweight ITAD orchestration system that answers:

What happened to this exact device from the moment it entered our custody until its final reuse, resale, donation, parts harvest or recycling destination?